

ACT-PKR-5500 Hydraulic Set Permanent Packer

Technical Datasheet DS-PKR-5500 Rev B | Product family: ACT-PKR-5500

1. Product Overview

The ACT-PKR-5500 is a compact hydraulic set permanent packer for 5-1/2 in. casing, designed for single- or dual-zone production completions and long-term injection service. One-piece C-ring slips and a bonded three-element package deliver dependable pack-off in L80 through P110 casing. The packer is removed by milling; an optional millout extension speeds retrieval.

2. Applications

- Permanent production completions
- Gas storage and injection wells
- Sour service completions (13Cr mandrel standard)
- Zonal isolation below ESPs

3. Features

- Hydraulic set - 10 brass shear screws, 45,000 lbf nominal set force
- One-piece C-ring slips, induction hardened wickers
- 13Cr mandrel standard for sour service (NACE MR0175)
- Bonded 3-element HNBR pack-off, 95 duro center
- Anti-swab feature prevents premature set while running
- API 11D1 Grade V0 qualified (with premium connections)

4. Technical Specifications

Parameter	Specification
Casing Size	5-1/2 in., 15.5 - 23 lb/ft
Maximum Tool OD	4.500 in.
Minimum Bore ID	2.375 in.
Overall Length	46.60 in.
Differential Pressure Rating	10,000 psi
Temperature Rating	325 degF
Setting Pressure	3,400 +/- 200 psi (10 brass screws)
End Connections	2-7/8 in. EUE 8RD (premium optional)
Tensile Rating (body)	220,000 lbf
Qualification	API 11D1 Grade V0 (premium conn.) / V3 (EUE)
Service	Sour (NACE MR0175) standard

All ratings are at ambient temperature unless otherwise noted. Performance envelope per API 11D1 V3 qualification testing where indicated. Consult ACT Engineering for combined load conditions.

5. Materials of Construction

Component	Material
Mandrel	API 5CT L80-13Cr, heat traceable
Cones / Piston	AISI 4140, 28-32 HRC
Slips	Ductile iron 80-55-06, wickers 50 HRC min
Element	HNBR 80/95/80
Shear screws	Brass C36000
O-rings	Viton 90A

6. Ordering & Reference Documents

Order by part number and specify: casing size and weight, service trim (standard or NACE sour), elastomer option, end connections, and any customer inspection requirements. Related documents:

- BOM-PKR-5500 Rev B
- ARP-PKR-5500 Assembly & Running Procedure
- TS-100
- MTR heat K73209 (L80-13Cr)

Datasheet DS-PKR-5500 Rev B issued 2024-11-05. Prepared by R. Callahan; approved by J. Martinez, P.E.. Specifications subject to change; confirm with ACT Engineering for well-specific designs.